5 **(a)** Explain how line spectra, together with the concept of photon, provide evidence for discrete energy levels in atoms.

.....

.....

..... [2]

**(b)** A discharge lamp contains excited atoms which emit electromagnetic radiation at several wavelengths. Figure 5.1 shows the three lowest electron energy levels of one of these atoms, labeled $n$ = 1, 2 and 3.

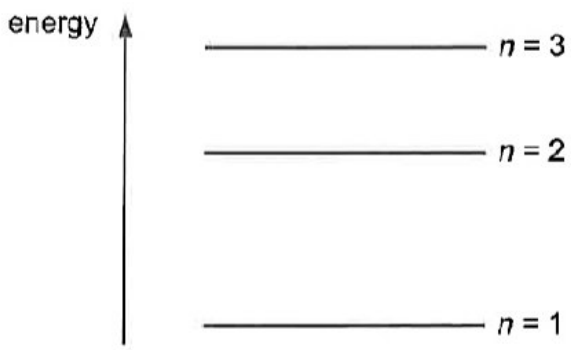



**Figure 5.1**

Electron transitions between these three energy levels can produce three different wavelengths of radiation. The transition from $n$ = 2 to $n$ = 1 produces radiation of wavelength 440 nm. Radiation of wavelength 590 nm is also emitted.

**(i)** Show that wavelength of the radiation emitted from the third transition is 250 nm.

[2]

**(ii)** State the region of the electromagnetic spectrum in which the third wavelength occurs.

..... [1]

[Total: 5]